BEYOND EVALUATION USE

An Empirical Study of Involvement and Use/Influence in Multi-Site Evaluations

Report 11:
Survey Data Report for the
Advanced Technological Education (ATE)
Program Evaluation

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Introduction

In November 2004, the National Science Foundation (NSF) awarded Dr. Frances Lawrenz and Dr. Jean King of the University of Minnesota a three-year grant to study the effect that participation in program evaluations has on the subsequent use of evaluation results. This grant, entitled "Beyond Evaluation Use", examined multiple projects within each of four main NSF-funded programs. These programs included: Local Systemic Change (LSC), Collaboratives for Excellence in Teacher Preparation (CETP), Advanced Technological Education (ATE), and Mathematics and Science Partnerships (MSP).

The grant activities were aimed at measuring and describing how these four programs carried out their multi-site evaluations. In particular, the research question was how these evaluations involved the various projects in this process. A further question focused on whether involvement affected the ways in which projects, individuals involved with the projects, and the STEM education and evaluation communities used the results.

As part of the research project, an on-line survey addressing the program-level evaluations was administered with evaluators and project leaders. The purpose of that *first* survey was to investigate the interaction of use and involvement and the data from that survey has been reported in "Report 1: NSF Follow-up Survey Detailed Data Report." Detailed case studies for each of the four programs (LSC, CEPT, ATE and MSP-RETA), which examine each program's history and evaluation context, and discuss survey findings within the context of each program can be found in Reports 2 – 5.[1] In particular, "Report 5: A Case Study of the Impact of the Advanced Technological Education (ATE) Program Evaluation" details and provides an analysis of the *first* survey data specific to the ATE Program.

After the administration of the *first* survey, the Beyond Evaluation Use team decided that further research into the relationship between evaluation use and influence should be done. The team re-administered the survey to gather more data for a confirmatory factor analysis. As with the *first* survey, the *second* survey was sent to PIs, but also included co-PIs. This report summarizes the results of a *second* administration of the original survey of ATE, LSC, CEPT and MSP project leaders. The purpose of this second administration was to gather data for a confirmatory analysis. Since this data was available, however, a presentation and analysis of the results is being given here. As was done with the first survey, an analysis of the results is given to help explain the extent to which ATE projects believed they were involved in the planning and implementation of the evaluation, and how that involvement affected the use of the program findings.

Along with this summary of the *second* administration of this survey, this report will also include selected results from the *first* administration of the ATE project leaders' survey. This is done to highlight both noteworthy differences that exist between these two administrations, as well as some of the consistencies that can also be seen.

The primary research questions guiding this inquiry are as follows:

[1] http://www.cehd.umn.edu/projects/beu/ . See "Project Documents", Reports 2-5.

1. What is the relationship between involvement in the ATE program evaluation and the use and influence of the evaluation on project-level staff and evaluators?
2. What factors other than involvement appear to be related to the use and influence of the program evaluation on the projects?

Methods

Data were collected through an online survey distributed to ATE project leaders. Additional data were collected through review of archival documents, publications, and reports produced by the ATE program and program evaluation, and through reflective descriptions provided by the ATE program evaluation principal investigator.

ATE Project Leader Survey

The Beyond Evaluation Use team acquired names and email addresses for the principal investigators (PIs) and co-principal investigators (co-PIs) from the NSF website. Although the lists contained the most recent email addresses on record, some names were missing email addresses. In such instances, a member of the EUG team searched the Internet for an email address.

The PIs and co-PIs received an initial invitation email with a description of the survey, an explanation of its use, the Human Subjects approval number, and telephone numbers to call with questions. The email also contained a link to the online survey with a login and password. Two and five weeks following the first email, the team sent email reminders to non-respondents. The reminder email also contained a link to the online survey with a login and password. The data were collected from April 8, 2009 through May 31, 2009.

The second survey data is available in "Report 9: NSF Second Follow-up Survey Detailed Data Report". Survey findings presented in this report are taken from 57 close-ended items and four open-ended questions related to the involvement in and use of the ATE program evaluation. There was also an opportunity for respondents to provide additional comments in a final open-ended item.

A pre-notice email was sent to approximately half (189 of the 381) of potential respondents, while the other half (192 of the 381) received no pre-notice email. Based upon response rates, the research team found that response rates were higher for those respondents who had received a pre-notice email (47%) than for those that did not (39%).

Due to the fact that many of the email addresses were obtained via Internet search by the EUG team, only approximately 10 percent of the emails were returned as undeliverable.
After receiving the initial email invitation, a number of potential respondents replied with valid reasons they should not be included in the survey, such as those on projects that had not yet begun the evaluation process or those who retired prior to their project being evaluated. These persons were removed from the sample. In total, 52 of the original 433 individuals contacted were either removed from the survey or had emails returned as undeliverable. Of these, 163 individuals replied resulting in a 44% response rate.

Team members reviewed responses to open-ended items. When appropriate, responses were used to support or exemplify findings from the close-ended survey items. The following table shows the number and percentage of respondents who provided comments to each open-ended question or statement.

Table 1 *Number of Responses to Open-ended Items*

Items 59, 60, 64	Comment	No Comment	Percent Commenting
What major factors, if any, enabled the evaluation experience to influence you?	58	105	36
What major factors, if any, limited the influence the evaluation experience had on you?	45	118	28
Is there anything else you would like to share with us about your experiences with the program evaluation?	39	124	24

Nonrespondent Study. To check for nonresponse bias, the team conducted a nonrespondent follow-up survey subsequent to the web survey of the PIs and co-PIs. Non-response error results when the people who responded to the survey are different from the individuals in the sample who did not respond, in a way that is relevant to the study. We developed a three-question nonrespondent survey to address the topics of involvement and use, as well as to gather some information about what kept individuals from responding to the initial survey and reminder emails. The nonrespondent survey items were:

1. How involved were you in the ATE Program Evaluation?
2. How much impact did the ATE Program Evaluation have on you?
3. What was the main thing that kept you from responding to our initial survey request?

A random sample of nonrespondents to the survey was drawn using Microsoft Excel. Each person was sent a personal, individually addressed email with a request for his or her input on the three items. Two reminder emails were sent to each of the respondents who did not respond to the initial request. Then, if email contact did not solicit a response, the individual was contacted by telephone. Thirty-two percent of the nonrespondents (24/74) completed the nonrespondent survey.

The results of the analysis comparing the ATE respondents and nonrespondents showed differences in the overall levels of involvement and use reported. The Involvement Mean of the survey respondents was higher (2.60) than the mean of the non-respondents (1.56). Similarly, the Use/Impact Mean was higher (2.66) for respondents than for non-respondents (1.56). The Involvement Mean for the respondents was calculated as the mean of thirteen questions on the original survey. The Use/Impact Mean was calculated in the same way, using the fourteen survey questions related to evaluation use and impact. The means for the non-respondents were simply the means of their answers to the two questions they responded to via email. Though these measurements are not identical, the two sets of means closely mirror one another in that the 4-point scale for each question or set of questions is the same.

During the analysis of *first* administration of the survey, team members sought to explain the difference that was seen between respondents and nonrespondents level of involvement reported.

At that time, team members hypothesize that the choice of the word "impact" as a global concept to encompass questions about use of the process, instruments, and results of the evaluation may have been responsible for this disparity. As a result, the difference between respondents and nonrespondents may be interpreted in one of two ways: as an indication of the possibility of upward bias in the estimates of overall use for ATE survey respondents, or as an artifact of the difference between use, influence and overall impact. (For more details, see "Report 5: A Case Study of the Impact of the Advanced Technological Education (ATE) Program Evaluation", p. 6.)

Archival Review

The collection of archival documents for ATE was conducted primarily using electronic databases and search engines. The NSF website provided information on the ATE program as a whole, as well as on individual ATE projects. The archival documents included reports, grant products, survey instruments, journal articles, NSF grant solicitations, project summaries, and newsletters. This review was conducted for the *first* administration of the survey. For details of that archival review, the reader is referred to Report 5. Details of the ATE program have been updated from Report 5 for purposes of this report to reflect current data concerning the program.

ATE Program and Evaluation Background and Context

ATE Program

ATE Program Facts

Period: 1993-present

Purpose: (1) Build capacity to provide advance tech education; (2) Increase number of skilled technicians.

Primary Target: Students & teachers in 2-year colleges and collaborating institutions.

Working under a Congressional mandate, the National Science Foundation (NSF) created the Advanced Technological Education (ATE) Program in 1992 to improve advanced technological education. The first award made by the ATE program was in October 1993. NSF aimed to increase the number and quality of skilled technicians in the workforce[2] thus improving U.S. competitiveness amidst growing global competition.[3] Viewed as a conduit for preparing technicians, community colleges received most of the ATE funds to strengthen scientific/technical education and training capabilities in biotechnology, environmental technology, GIS, manufacturing, and telecommunications.[4]

ATE Program Evaluation Background

Between 1994 and 2005, ATE funded 674 projects and centers, making grants to 345 unique institutions of which more than 200 were two-year colleges. Total cost of the program was approximately $350 million. ATE funded large Centers of Excellence ($800,000 for 3 years) and smaller-scale projects (approximately $150,000 for 3 years). ATE projects focused on improving technical education materials, enhancing technical instruction, and providing professional development to faculty and teachers. Grantees established partnerships with high schools, two-

[2] ATE Final Report, 2006, p. vi
[3] Gullickson, Lawrenz, & Keiser, 2000.
[4] ATE Final Report, 2006, p. 33.

and four-year colleges, businesses, government agencies, and professional societies to improve the nation's advanced technology workforce.[5,6] ATE centers could not be purely local; they required either a regional or national focus.[7] In 1999, NSF funded the Evaluation Center at Western Michigan University (WMU) to conduct an external evaluation of the entire ATE program to collect data for ATE program reporting and to offer program improvement feedback to individual ATE projects.[8] Arlen Gullickson, Principal Investigator, received two consecutive NSF grants: the initial grant of $1.3 million covered 1999-2003, and a subsequent $1.8 million grant covered 2002-2006.[9] Since that time, two grants have been awarded to Gullickson to continue with ATE program evaluations.

ATE Evaluation Facts

NSF Grant Funding
•$1.3 million (1999 – 2003)
•$1.8 million (2002 – 2006)
•$.6 million (2006 – 2009)
•$1.04 million (2008 – 2010)

NSF Goals for ATE Program Evaluation
•Provide evidence of quality of ATE program
•Inform program improvement

Evaluation Questions
•To what degree is the program achieving its goals?
•Is it making an impact and reaching the individuals and groups intended?
•How effective is it when it reaches its constituents?
•Are there ways to significantly improve the program?

Personnel/Expertise
•NSF 3-person mgmt team
•Experienced evaluation team
•Eval Advisory Committee

To meet the reporting requirements of the Government Performance Results Act (GPRA), the ATE evaluation design focused on being "responsive to and meeting the needs of NSF." However, part way through the multi-year ATE program evaluation, NSF shifted its focus to highlight the need for evaluation products to be useful to STEM educators and evaluators.[10]

ATE Program Evaluation Questions

The evaluation assessed the impact and effectiveness of the ATE program by addressing four basic evaluation questions:

1. To what degree is the program achieving its goals?
2. Is it making an impact and reaching the individuals and groups intended?
3. How effective is it when it reaches its constituents?
4. Are there ways the program can be significantly improved?[11]

ATE Program Evaluation Design

Principal evaluation activities included: collecting descriptive data, developing a web-based evaluation survey to collect data over time, conducting site visits, drafting targeted reports on

[5] The Sustainability of Advanced Technological Education Supported Efforts: An Evaluation Wayne W. Welch, University of Minnesota (ret.) and Arlen R. Gullickson, Western Michigan University August 2006, p.2, accessed htttp://www.wmich.edu/evalctr/ate/ATESustainabilityReport.pdf
[6] ATE Program Evaluation Project. Presentation accessed WMU website at http://www.wmich.edu/evalctr/ate/presentations/gettingstartedfinal.pdf
[7] NSF, 03 Program Solicitation, p. 2
[8] ATE Program Evaluation Project. Presentation accessed WMU website at http://www.wmich.edu/evalctr/ate/presentations/gettingstartedfinal.pdf
[9] ATE Final Report, 2006.
[10] ATE Final Report, 2006, p. 16.
[11] http://www.wmich.edu/evalctr/ate/question.html

various aspects of the program, and providing formative evaluation information to NSF and to the projects and centers.[12]

The ATE evaluation featured annual web-based surveys and multiple site visits for gathering primarily qualitative information. The annual surveys collected quantitative data on the activities, accomplishments, and effectiveness of ATE projects and centers for general accountability purposes; the site visits yielded data on the underlying drivers of program success including collaboration with partners, professional development, and project sustainability. The evaluation also included targeted studies on various aspects of the program and formative evaluation information to NSF and to the projects and centers.

ATE Program Evaluation Components

The ATE program evaluation contained the following components:

- *Annual surveys*. Web-based annual grantee surveys provided common information across all sites about the program operations, the numbers and types of participants, and project activities and interactions.[13]
- *Site visits.* Teams of experts visited 13 ATE sites and used prepared protocols to interview PIs, industry collaborators, faculty and students.
- *Site visit guide.* Following the visits to 13 sites, the ATE evaluation team created a detailed manual or handbook to aid ATE projects and centers in how to conduct site visits.[14]
- *Issue papers.* A series of issue papers synthesized the survey data, site visit data, and existing research.[15]
- *Targeted studies*. In-depth targeted studies on various aspects of the program were conducted.[16]
- *Dissemination.* The prolific ATE evaluation team produced more than 50 reports and publications which are archived at the WMU Evaluation Center website at http://www.wmich.edu/evalct.
- *National Visiting Committee Handbook.* Site visits by WMU evaluators resulted in the production of the *National Visiting Committee Handbook* to help those responsible for organizing an NVC, planning and conducting meetings, and for post-meeting activities.

- *Brochures.* To meet NSF's goal of wide dissemination of the ATE evaluation data, the evaluation team produced tri-fold brochures to better reach ATE collaborators, administration, and other supporters.

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[12] ATE Program Evaluation Project. Presentation accessed WMU website at http://www.wmich.edu/evalctr/ate/presentations/gettingstartedfinal.pdf

[13] ATE Final Report, 2006, p. 14.

[14] A Guide for Planning and Implementing Site Visits, Frances Lawrenz, University of Minnesota, Nanette Keiser, Western Michigan University, and Beth Lavoie, University of Minnesota. March 2002.

[15] The ATE Program: Issues for Consideration by Frances Lawrenz and Nanette Keiser, Winter 2002, p.9. http://www.wmich.edu/evalctr/ate/ATEpapers/issuesforconsider.pdf

[16] ATE Program Evaluation Products. The Evaluation Center, Western Michigan University. Accessed 2/15/07 at http://www.wmich.edu/evalctr/ate/evalproducts.htm.

- *Meta-Evaluation.* Four external meta-evaluators observed and analyzed the WMU evaluation activities. [17]

Project Leader Involvement and Use

This section presents data from the survey to describe how project leaders (i.e., PIs and co-PIs) reported being involved in and influenced by the ATE program evaluation. This presentation of findings is organized around six topics that guided the data collection and analysis for the study. For each topic, quantitative results are discussed. The six topics are:

- Motivation for Involvement
- Levels of Involvement
- Perceived Effect of Involvement
- Use by Project Leaders and Evaluators
- Use by Projects
- Factors Affecting Use

Motivation for Involvement

Item 4 of the survey addressed the motivation that PIs and co-PIs had, at different stages (planning, implementation and communication) of the evaluation, for being involved. The planning stage includes activities such as discussing the focus of the evaluation, identifying evaluation planning team members, developing the evaluation plan, while the implementation stage includes developing data collection instruments and processes, collecting, analyzing, reviewing, and interpreting data. The final stage of the program evaluation – the communication stage – is comprised of activities such as developing future plans for your project, writing and reviewing evaluation reports, presenting evaluation findings. Of the three stages of the evaluation, findings suggest that motivation for involvement was always nearly equally distributed among being required, being asked and volunteering. One exception to this finding might be a small variation in the communication stage of the evaluation where we see that project leaders were slightly more likely to report being motivated by a requirement (36.8%), than by either volunteering (34.6%) or being asked (28.6%). This, however, may be a reflection of the fact that NSF requires dissemination of the evaluation findings, and thus PIs may have thought that communication of findings was being required of them.

Table 2 *Motivation for Involvement*

Item 4	I was required	I was asked	I volunteered	N
Planning stage	33.1%	34.6%	32.3%	127
Implementation stage	33.6%	34.3%	32.1%	134
Communication of evaluation findings stage	36.8%	28.6%	34.6%	133

[17] Hartmann and Loizides (2001), Migotsky and Stake (2001), Pucel (2003), Mark (2006).

Levels of Involvement

The survey asked respondents about their perceptions of involvement during the planning, implementation and communication phases of the program evaluation. Responses to the involvement questions are summarized in Tables 3, 4 and 5. Each involvement question began with "I was involved in…."

Table 3 *Involvement – Planning*

Items 1-3	No	Yes, A little	Yes, Some	Yes, Extensively	N	*M*	*SD*
Discussions that focused the evaluation	14.6%	8.9%	24.8%	51.6%	157	3.13	1.09
Identifying evaluation planning team members	29.9%	9.7%	18.2%	42.2%	154	2.73	1.29
Developing the evaluation plan	26.5%	10.3%	28.4%	34.8%	155	2.72	1.20

Table 4 *Involvement – Implementation*

Items 5 - 10	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
Developing data collection instruments.	34.0%	17.3%	23.1%	25.6%	156	2.40	1.20
Developing data collection processes.	29.0%	18.1%	25.2%	27.7%	155	2.52	1.18
Collecting data.	33.8%	15.9%	18.5%	31.8%	157	2.48	1.25
Reviewing collected data for accuracy and/or completeness.	37.6%	8.3%	21.0%	33.1%	157	2.50	1.29
Analyzing data.	38.7%	11.0%	27.7%	22.6%	155	2.34	1.21
Interpreting collected data.	29.9%	11.0%	29.9%	29.2%	154	2.58	1.20

Table 5 *Involvement – Communication*

Items 12 - 15	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
Writing evaluation reports.	50.0%	15.6%	16.2%	18.2%	154	2.03	1.18
Reviewing evaluation reports for accuracy and/or completeness.	23.5%	17.0%	24.8%	34.6%	153	2.71	1.18

Presenting evaluation findings.	33.3%	7.8%	22.9%	35.9%	153	2.61	1.28
Developing future project plans based on evaluation results.	16.6%	9.9%	23.8%	49.7%	151	3.07	1.12

Of these three phases, respondents indicated the most extensive overall involvement in the planning stage. During this phase, more than half of all respondents (51.6%) felt extensively involved in discussions that focused the evaluation. However, respondents also felt extensively involved in the other two planning phase activities: identifying evaluation planning team members (42.2%) and developing the evaluation plan (34.8%).

It is also notable that results from the *first* administration of the survey indicate a much larger proportion of respondents indicating no involvement during the planning stage than respondents to the *second* survey. (See Table 3A for results of the first survey.) At that time, over a third of all respondents claimed no involvement in each activity during this stage (37% were not involve in discussions, 48% were not involved in identifying a planning team, and 41% were not involved in developing the plan.) A corresponding rise in respondents involved `some' can also be seen. One reason for this change might be that the main component of the ATE program evaluation – the yearly survey – has been changing in response to input from ATE PIs. As a result of PIs feedback, more involvement can be seen.

Table 3A *Involvement – Planning (First Survey)*

Items 1-3	No	Yes, a little	Yes, some	Yes, extensively	N	Mean	SD
Discussions that focused the evaluation.	37.1%	5.1%	10.1%	47.8%	178	2.7	1.4
Identifying evaluation planning team members.	47.7%	5.2%	9.8%	37.4%	174	2.4	1.4
Developing the evaluation plan.	41.2%	9.0%	14.7%	35.0%	177	2.4	1.3

During the communication phase of the evaluation, nearly half of all respondents perceived themselves as being extensively involved with the development of future projects plans based on the evaluation results. This may be due to a couple of factors. First, many ATE projects received further funding to encourage the use of what had been learned through the evaluation process. Thus, this further funding served as a catalyst for the extensive involvement we see reported in this aspect of the communication stage. Further, since the NSF requires that PIs explain their success and failures in past projects in current proposals, evaluation results would necessarily play a role in the future project plans detailed in these proposals. In both cases, involvement in the communication stage is being motivated, in part, by outside forces.

While half of all respondents felt no involvement in writing the evaluation reports, they nevertheless did see themselves as involved with reviewing evaluation reports and presenting evaluation findings.

Overall, PIs and co-PIs characterize the evaluation planning stage as that stage of the evaluation where they were most extensively involved. Moving into the implementation stage saw an increase in the number of respondents perceiving themselves as less involved. The communication stage reversed this trend, but only minimally. A similar trend is seen in the results of the *first* survey.

Perceived Effect of Involvement

Questions 17 through 22 addressed respondents' perceptions concerning the influence their input had on each stage of the program evaluation. In providing this feedback, respondents were asked to differentiate between having their input taken into consideration and having their input actually used. Survey results suggest that PIs and co-PIs perceived their input as being both considered as well as used during each stage of the evaluation. That is to say that input use was not perceived differently from input consideration at any stage of the evaluation, suggesting that respondents' judged their input to be influential overall. Moreover, the degree to which they viewed their influence as either considered or used was also similar across the stages of the evaluation. Between 35 – 40% of respondents said that their input was both considered and used extensively in the evaluation, while another 24 – 30% claimed that their input was considered or used "some".

Influence was also similarly rated during the *first* survey (see Table 6A). There, however, we saw respondents reporting even higher rates of having the input considered and used "extensively" (from 49 – 56%), and slightly higher rates of having their input considered or used "some" 31 – 34%).

Table 6 *Influence on the Evaluation*

Items 17-22	Was not involved	No	Yes, a little	Yes, some	Yes, extensively	N
Input during *planning* was taken into consideration	18.0%	.6%	6.8%	30.4%	40.4%	161
Input during *implementation* was taken into consideration	17.5%	.0%	8.8%	28.8%	38.8%	160
Input during *communication* was taken into consideration	16.5%	1.3%	13.3%	26.6%	38.0%	158
Input during *planning* was used.	17.1%	1.3%	12.0%	28.5%	35.4%	158
Input during *implementation* was used.	16.7%	.6%	11.5%	28.2%	34.6%	156
Input during *communication* was used.	15.4%	.6%	14.1%	24.4%	38.5%	156

Table 6A *Influence on the Evaluation (First Survey)*

Items 17-22	No	Yes, a little	Yes, some	Yes, extensively	N*

Input during *planning* was taken into consideration	1.7%	11.1%	31.6%	55.6%	117
Input during *implementation* was taken into consideration	3.4%	12.7%	31.4%	52.5%	118
Input during *communication* was taken into consideration	2.6%	12.2%	32.2%	53.0%	115
Input during *planning* was used.	1.8%	15.3%	32.4%	50.5%	111
Input during *implementation* was used.	1.8%	15.5%	33.6%	49.1%	110
Input during *communication* was used.	4.6%	11.1%	32.4%	51.9%	108

Perceived Use by Project Leaders

The survey asked project leaders to reflect on the extent to which the program evaluation influenced their knowledge/understanding, skills and beliefs. Additionally, project PIs were asked about their use of the program evaluation process and findings in another evaluation context. Each of these is examined in turn.

Influence on Knowledge and Understanding

Respondents most commonly believed that their knowledge and understanding of evaluation (all stages), STEM education and evaluation, and their own project increased "some" as a result of the program evaluation (see Table 7). While the program evaluation appears to have most strongly influenced respondents' knowledge and understanding of how to plan an evaluation, evaluation implementation and communication were also influenced for the majority of project leaders. With respect to STEM education and evaluation, respondents reported that the program evaluation was less likely to increase knowledge and understanding of STEM education (M = 2.41, SD = 0.99) than STEM education evaluation (M = 2.59, SD = 0.91). Nearly 23% of respondents said that the program evaluation had no influence on their knowledge and understanding of STEM education. However, when looking at the "no" influence category, response rate is notably lower than those seen in the first survey administration (see Tablee 7A). There, between 21% and 32% of respondents claimed no influence on knowledge and understanding in any of the stages of the evaluation, or on STEM education, STEM evaluation or their own projects.

Finally, over 70% of respondents believed their knowledge and understanding of their own projects was increased as a result of the program evaluation. As one respondent wrote "The process of developing the evaluation made me think about the project in a different and productive manner."

Table 7 *Influence on Knowledge and Understanding*

Items 25-30	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
The evaluation increased my knowledge/ understanding of how to plan an evaluation.	10.9%	23.7%	44.9%	20.5%	156	2.75	0.91
The evaluation increased my knowledge/ understanding of how to implement an evaluation.	11.0%	25.8%	43.9%	19.4%	155	2.72	0.90
The evaluation increased my knowledge/ understanding of how to communicate evaluation findings.	12.3%	25.3%	42.2%	20.1%	154	2.70	0.93
The evaluation increased my knowledge /understanding of STEM education.	22.7%	27.9%	35.1%	14.3%	154	2.41	0.99
The evaluation increased my knowledge /understanding of STEM education evaluation.	14.2%	27.7%	43.2%	14.8%	155	2.59	0.91
The evaluation increased my knowledge/ understanding of my project.	11.7%	18.2%	39.0%	31.2%	154	2.90	0.98

Table 7A *Influence on Knowledge and Understanding (First Survey)*

Items 25-30	No	Yes, a little	Yes, some	Yes, extensively	N	M	SD
The evaluation increased my knowledge/ understanding of how to plan an evaluation.	22.3%	25.0%	34.2%	18.5%	184	2.5	1.0
The evaluation increased my knowledge/ understanding of how to implement an evaluation.	21.2%	27.7%	34.8%	16.3%	184	2.5	1.0
The evaluation increased my knowledge/ understanding of how to communicate evaluation findings.	25.7%	24.6%	35.5%	14.2%	183	2.4	1.0
The evaluation increased my knowledge /understanding of STEM education.	31.9%	22.0%	31.9%	14.3%	182	2.3	1.1
The evaluation increased my knowledge /understanding of STEM education evaluation.	23.2%	22.7%	37.6%	16.6%	181	2.5	1.0
The evaluation increased my knowledge/ understanding of my project.	22.1%	18.2%	35.9%	23.8%	181	2.6	1.1

Influence on Skills

Although the program evaluation's influence on skills reflects a similar pattern to its influence on knowledge and understanding, slightly higher percentages of respondents reported no influence (see Table 8). The pattern was also similar to those found for influence on knowledge and understanding in that STEM education was the area with the least amount of influence as indicated by lowest mean and highest percentage reporting no influence. Further, respondents reported that their evaluation planning skills were those most influenced of all of the stages of evaluation.

Table 8 *Influence on Skills*

Items 31-36	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*

The evaluation improved my skills in planning an evaluation.	16.1%	18.7%	46.5%	18.7%	155	2.68	0.96
The evaluation improved my skills in implementing an evaluation.	14.8%	23.9%	43.9%	17.4%	155	2.64	0.94
The evaluation improved my skills in communicating evaluation findings.	14.2%	27.7%	40.6%	17.4%	155	2.61	0.94
The evaluation improved my skills as a STEM educator.	22.6%	28.4%	34.2%	14.8%	155	2.41	1.00
The evaluation improved my skills as a STEM education evaluator.	19.0%	28.8%	37.3%	15.0%	153	2.48	0.97
The evaluation improved my skills for working on my project.	14.8%	23.9%	38.7%	22.6%	155	2.69	0.98

Some of the specific skills that respondents noted as examples of skills improved by the program evaluation were that "it helped with budget considerations" and "collecting data". Respondents frequently noted that their experience allowed them "to refine and broaden my evaluation skills and apply them to other projects…". Further, skills needed for working on one's own project were often identified as improved by the evaluation. As one respondent wrote, "the evaluation experience has influenced me by allowing me to understand the many different aspects of evaluation … and how complex, yet important, a relevant evaluation of the project's performance is to adjust[ing] the project so it can meet its objective and goals."

Influence on Beliefs

In much the way that respondents were asked to reflect upon the program evaluation's influence on their knowledge, understanding and skills, respondents were asked to further reflect upon the influence that the evaluation had on their *beliefs* about the importance of the evaluation. Table 9 presents these results. Once again we see a pattern that mirrors the influence respondents reported for their knowledge and understanding as well as for their skills. One interesting difference in the pattern noted thus far lies in the fact that the communication stage of the evaluation was the stage identified as having the greatest increase as far as importance was concerned. Given that this stage had the least influence on knowledge and understanding, as well as on skills, one suggestion to explain the increase in its significance may be that having to communicate findings made respondents more invested in what they are reporting. Another possible suggestion is that PIs became increasingly aware of the importance of communication since they, themselves, wanted more communication. One further observation is the relatively larger influence on respondents' belief about the importance of program evaluation than on either STEM education or STEM education evaluation, or the importance of an individual's project. Perhaps this can best be explained as more of a reflection on respondents' previously held beliefs concerning the importance of STEM education (thus, the evaluation had little effect on increasing that), while their beliefs about the importance of evaluation, itself, were more likely to increase.

Table 9 *Influence on Beliefs*

Items 38-43	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
The evaluation increased my belief in the importance of planning an evaluation	12.2%	17.3%	41.7%	28.8%	156	2.87	0.97
The evaluation increased my belief in the importance of implementing an evaluation	12.8%	21.2%	36.5%	29.5%	156	2.83	1.00
The evaluation increased my belief in the importance of communicating evaluation findings	13.1%	16.3%	45.8%	24.8%	153	2.82	0.95
The evaluation increased my belief in the importance of STEM education.	27.7%	25.2%	27.1%	20.0%	155	2.39	1.10
The evaluation increased my belief in the importance of STEM education evaluation.	19.5%	25.3%	31.8%	23.4%	154	2.59	1.05
The evaluation increased my belief in the importance of my project.	16.9%	20.1%	37.7%	25.3%	154	2.71	1.03

With respect to questions 25-43, all of which concern some sort of influence that the evaluation had, it is important to note that this influence may have been a negative influence. For example, the program evaluation may have failed to give adequate advance notification for the completion of a task and, as a result, the respondent learned that it was important to do it differently. To address issues of this sort, question 37 asked whether the evaluation had a negative impact and then offered space for a written explanation. Only 6% of survey participants responded to this question (10/163). With so few responses, there were no trends identified. Some responses from this question include "The data sought did not add value", "The initial results and report of the evaluators was exceedingly academic and not really focused on the project" and "I found the evaluation questions off target for the goals of our project".

Use in another Context

Questions 45 – 49 sought information from respondents concerning the use of the evaluation knowledge, products, or processes in another (subsequent) evaluation context. The survey directed only respondents who had participated in a subsequent evaluation to respond to these questions. If a respondent had not participated in another evaluation, they were instructed to continue with question 50.

For those respondents who participated in another evaluation after the ATE program evaluations, over 80% used what they learned from planning or from implementing the evaluation in a subsequent evaluation context. Although they reported a high rate of using what they learned from these two stages, respondents, nevertheless, were not as likely to use the actual evaluation plan from the program evaluation. Over 1/3 of respondents said they did not use the plan at all as a model for a future evaluation. But even the least-used aspect of the evaluation, the data collection instruments (with 36.6% reporting that they did not use the data collection instruments in another evaluation), still had over 60% of respondents using those instruments at least "a little" in another evaluation subsequent to the ATE program evaluation. Given that another

evaluation would have its own unique context and, thus, most likely require its own plan and data collection instruments, the interesting point to note is the actual level of usage that does occur. One hypothesis is that the plan and data collection instruments act as models or templates for future evaluations.

Table 10 *Use of Knowledge, Products, Processes in Another Evaluation Context*

Items 45-49	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*
I used what I learned from planning the program evaluation in another evaluation.	12.2%	24.4%	43.9%	19.5%	41	2.71	0.93
I used the evaluation plan from the program evaluation as a model in another evaluation.	34.1%	22.0%	29.3%	14.6%	41	2.24	1.09
I used what I learned from implementing the program evaluation in another evaluation.	17.1%	29.3%	36.6%	17.1%	41	2.54	0.98
I used data collection instruments from the program evaluation in another evaluation.	36.6%	22.0%	29.3%	12.2%	41	2.17	1.07
I used what I learned from communicating evaluation findings from the program evaluation in another evaluation.	26.8%	24.4%	31.7%	17.1%	41	2.39	0.91

Use of Findings

Questions 50 – 58 asked PIs and co-PIs to report the ways in which they used the program evaluation findings. Table 11 reports these results. One notable outcome of this series of questions is the large percentage of respondents using the program evaluation findings to develop future plans for their projects (35.3% used them "extensively", with another 36.5% using them "some"). Respondents also used the program evaluation finding to increase the attention given to their projects (M = 2.73, SD = 1.04) and to make changes to their projects (M = 2.71, SD = 1.01). On the other hand, using the program evaluation finding to promote additional evaluation of their projects was the least likely use of findings that respondents reported. Over 45% of respondents claimed that they did not use evaluation findings for this purpose.

In general, however, results from this *second* survey administration indicated a significantly smaller number of people claiming no use than in the *first* administration. At that time (see Table 11A), over half of respondent indicated that they made no use of the program evaluation to seek funding or to promote additional evaluation of their project. Other potential uses rated between 30 – 41%. One possible reason for this difference might be that in the early stages of the ATE program, project leaders were less aware of opportunities for using the findings of the evaluation than those in later years of the ATE program.

Table 11 *Use of Findings*

Items 50-58	No	Yes, a little	Yes, some	Yes, extensively	N	*M*	*SD*

I used the program evaluation findings to make decisions about the future existence of my project (e.g., cancel or continue).	25.0%	14.7%	32.1%	28.2%	156	2.63	1.14
I used the program evaluation findings to develop future plans for my project.	15.4%	12.8%	36.5%	35.3%	156	2.92	1.04
I used the program evaluation findings to increase the attention given to my project.	17.3%	19.2%	36.5%	26.9%	156	2.73	1.04
I used the program evaluation findings to meet contractual and/or legal requirements of my project.	25.2%	16.1%	32.9%	25.8%	155	2.59	1.13
I used the program evaluation findings to enhance my organization's commitment to my project.	28.4%	20.0%	33.5%	18.1%	155	2.41	1.09
I used the program evaluation findings to seek funding for my project(s).	31.6%	14.8%	31.0%	22.6%	155	2.45	1.16
I used the program evaluation findings to promote additional evaluation of my project(s).	45.2%	11.6%	27.1%	16.1%	155	2.14	1.17
I used the program evaluation findings to make changes to my project(s).	17.8%	15.9%	43.3%	22.9%	157	2.71	1.01
I used the program evaluation findings to help plan a new project(s).	24.5%	13.5%	36.8%	25.2%	155	2.63	1.11

Table 11A *Use of Findings (First Survey)*

Items 50-58	No	Yes, a little	Yes, some	Yes, extensively	N*	Mean	SD
I used the program evaluation findings to make decisions about the future existence of my project (e.g., cancel or continue).	39.7%	15.2%	30.5%	14.6%	151	2.2	1.1
I used the program evaluation findings to develop future plans for my project.	29.8%	19.9%	32.5%	17.9%	151	2.4	1.1
I used the program evaluation findings to increase the attention given to my project.	37.5%	17.1%	29.6%	15.8%	152	2.2	1.1
I used the program evaluation findings to meet contractual and/or legal requirements of my project.	35.8%	12.6%	26.5%	25.2%	151	2.4	1.2
I used the program evaluation findings to enhance my organization's commitment to my project.	40.7%	14.7%	30.7%	14.0%	150	2.2	1.1
I used the program evaluation findings to seek funding for my project(s).	54.7%	12.0%	20.7%	12.7%	150	1.9	1.1
I used the program evaluation findings to promote additional evaluation of my project(s).	55.0%	16.6%	19.9%	8.6%	151	1.8	1.0
I used the program evaluation findings to make changes to my project(s).	30.9%	23.0%	34.2%	11.8%	152	2.3	1.0
I used the program evaluation findings to help plan a new project(s).	36.8%	22.4%	26.3%	14.5%	152	2.2	1.1

Comparisons Among the Types of Influence

As shown in Table 12, data from the survey suggest that the ATE program evaluation had the greatest influence on project leaders' beliefs (M = 2.72, SD = 0.89). Nevertheless, the influence that the program evaluation had on knowledge and understanding, skills and ways of using findings was also notable. The only area where influence lagged (M = 2.41, SD = 0.91) was the subsequent use of what was learned from the program evaluation. Here, however, we must be careful interpreting results due to the small number of respondents (N = 41) in comparison with the other areas of influence.

Table 12 *Influence of program evaluation on project leaders*

Type of Influence	N	*M*	*SD*
Influence on knowledge/understanding (q. 25-30)	153	2.68	0.82
Influence on skills (q. 31-36)	152	2.59	0.88
Influence on beliefs (q. 38-43)	150	2.72	0.89
Use of knowledge, products, processes in other contexts (q. 45-49)	41	2.41	0.91
Ways of using findings (q. 50-58)	150	2.58	0.88

Use by Projects

As we saw above, survey items focused on the ways in which project leaders were influenced by the program evaluation, including the extent to which they used the process and findings to make decisions about projects. Some of the open-ended responses allowed for deeper understanding about the ways in which the program evaluation influenced the projects. (Open-ended items are listed in Table 1, along with the number of respondents commenting and the rate of commenting.) One important caveat concerning the open-ended items concerns the fact that some of the respondents answered these questions in reference to their own project's evaluation, as opposed to the larger program evaluation.

In spite of this limitation, the evaluation team examined open-ended items carefully with this possible misunderstanding in mind, and found that several themes emerged relating to the influence of the program-level evaluation on respondents.

One theme was the hope that the ATE evaluation could yield insights relevant to individual projects. "As a PI it would be helpful to see general ATE results and interpretation in order to improve my own project work". However another respondent noted that the evaluation alone might not provide those insights. "It was more of a census that feeds into a larger portrait of the overall program than something that necessarily feeds back into the project."

Another theme that emerged was a concern about the time commitment associated with the evaluation would detract from other activities. One respondent noted that the ATE evaluation is "extensive to the point of being obtrusive. Seems that I spend a lot of time completing evaluations in addition to my project's own evaluation." Another noted, "It is very difficult to

collect all of the data required to complete this study when you have a large number of entities involved in your center."

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Influence on the Field

Questions 61 and 62 looked at respondent's assessment of how influential the ATE program evaluation was on the STEM education community and the evaluation community. Tables 13 and 14 summarize these survey responses.

Table 13 *Influence on STEM Education Community*

Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
8.7%	28.7%	36.7%	17.3%	8.7%	150

Table 14 *Influence on the Evaluation Field*

Not influential at all	Marginally influential	Somewhat influential	Influential	Very influential	Total
5.9%	23.0%	36.8%	26.3%	7.9%	152

Overall, we can see that the majority respondents believed the program evaluation to be at least somewhat influential on both the STEM education community and the evaluation community. More than a quarter of respondents felt that the impact was either influential or very influential.

Factors Affecting Use

Survey respondents were asked to identify any major factors that either enabled or limited the influence of the evaluation on them. Fifty-eight out of 163 respondents (36%) provided feedback on enabling influences, while a smaller percentage (45 out of 163, or just 28%) supplied feedback to the limiting influences question.

Enabling Factors

In a broad sense, collaboration facilitated use among ATE project leaders. Survey respondents noted specifically that feedback from their peers provided a larger perspective for their projects. As one respondent wrote, this feedback "increased my awareness of where other evaluators placed their emphases, and pointed out some evaluation areas I need to further pursue…"

Respondents also noted that working with experienced external evaluators provided them with valuable feedback not only in terms of content, but also in terms of helping with time management and motivation. They identified, in particular, the "the high quality of the person charged with evaluation responsibilities." Along these lines, evaluators were identified as enabling factors because they "assisted in the interpretation and use of evaluation data", "provided concrete examples and suggestions that were easy to follow and implement" and a demonstrated a "willingness … to discuss the process and listen to my suggestions". In general,

involved, experienced, and skilled evaluators were had a positive effect on an evaluations influence.

Another enabling factor was the use of the ATE evaluation in educating others about the value of the respondent's project. "The evaluation findings and its communication helped me to enhance my organization's and NSF's commitment to my project. The evaluation results and reports helped to educate the administrators, the majority of whom do not have a STEM background, education, and teaching experience."

Limiting Factors

When it comes to factors that limited the influence of the evaluation on respondents, the already mentioned time constraints were paramount. Repeatedly, feelings of having "too little time to complete all of the evaluation components…" and having " limited time to spend on it due to my duties to manage every other aspect of our ATE project" were expressed. One respondent commented that, it "seemed to me at the time to be a short time to get the required data -- the next one will be easier now that I know what to do and will start on it earlier."

Aside from time limitations, respondents also were concerned that the survey was not clear:

> Many of the categories were so ambiguous. I felt it was hard to answer correctly. I wish there was a sample survey session to go through at ATE in October so that first time PIs know how to prepare information for the survey and also so PIs that have done it could be in a session with other PIs to hear their issues with the survey. I believe this would help give some consistency to how the survey is understood.

A significant concern for some participants was that they did not understand the purpose of the ATE evaluation. As one respondent wrote:

> My only recollection of [the ATE survey] is that we responded to a set of questions. Can't remember when. I don't recall this being a big planning event in our grant project, or calling for data that we did not already have. What I do remember was counting participants, that sort of thing. So the fact that I did not understand the implications or purpose of the ATE evaluation limited its influence on me. I am pro-evaluation, so would have welcomed meaningful discussion. I don't recall that any occurred in relation to this evaluation.

Summary

This document summarizes the ways in which ATE project leadership experienced involvement in the ATE program evaluation and how they subsequently used the results of this evaluation.

Of the three stages of the evaluation, findings suggest that project leader motivation for involvement was always nearly equally distributed among being required, being asked and volunteering. Overall, PIs and co-PIs characterize the evaluation planning stage as that stage of the evaluation where they were most extensively involved. Moving into the implementation

stage saw an increase in the number of respondents perceiving themselves as less involved. The communication stage reversed this trend somewhat. Most respondents said that their input was both considered and used either extensively or "some" in the evaluation.

Data from the survey suggest that the ATE program evaluation had the greatest influence on project leaders' beliefs. Nevertheless, the influence that the program evaluation had on knowledge and understanding, skills and ways of using findings was also notable. The only area where influence lagged was is the subsequent use of what was learned from the program evaluation. Overall, we can see that the majority of respondents believed the program evaluation to be at least somewhat influential on both the STEM education community and the evaluation community. More than a quarter of respondents felt that the impact was either influential or very influential.

One theme that emerged concerned the role of collaboration with peers in helping the projects. Project leaders also noted the importance of having a highly qualified, involved external evaluator available for continual feedback. This aligns well with survey respondents' identification of collaboration with colleagues and the evaluation team as a key common enabling factor. The main factor that limited influence was lack of time on the part of the respondents to accomplish all of the evaluation tasks. Lack of clarity of the survey and not understanding the purpose of the evaluation were also identified as factors limiting the evaluation influence.